

Practical 5

Aim: Demonstrate normalization, standardization, and data encoding on a sample dataset using Python.

Source-Code:

```
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import LabelEncoder
data = {
    'Name': ['Amit', 'Priya', 'Rahul', 'Sneha', 'Karan'],
    'Age': [22, 25, 28, 24, 30],
    'Salary': [25000, 30000, 35000, 28000, 40000],
    'Gender': ['Male', 'Female', 'Male', 'Female', 'Male'],
    'Department': ['HR', 'IT', 'Finance', 'IT', 'HR']
}
df = pd.DataFrame(data)
print("Original Dataset:")
print(df)
print("\n----- Normalization -----")
normalizer = MinMaxScaler()
df[['Age_Normalized', 'Salary_Normalized']] = normalizer.fit_transform(
    df[['Age', 'Salary']]
)
print(df[['Age', 'Age_Normalized',
           'Salary', 'Salary_Normalized']])
print("\n----- Standardization -----")
standardizer = StandardScaler()
df[['Age_Standardized', 'Salary_Standardized']] = standardizer.fit_transform(
    df[['Age', 'Salary']]
)
print(df[['Age', 'Age_Standardized',
           'Salary', 'Salary_Standardized']])

print("\n----- Label Encoding -----")
label_encoder = LabelEncoder()
df['Gender_Encoded'] = label_encoder.fit_transform(df['Gender'])
print(df[['Gender', 'Gender_Encoded']])
print("\n----- One-Hot Encoding -----")
department_encoded = pd.get_dummies(df['Department'])
print(department_encoded)
print("\n----- Final Dataset -----")
print(df)
```

Output:

Original Dataset:

	Name	Age	Salary	Gender	Department
0	Amit	22	25000	Male	HR
1	Priya	25	30000	Female	IT
2	Rahul	28	35000	Male	Finance
3	Sneha	24	28000	Female	IT
4	Karan	30	40000	Male	HR

----- Normalization -----

	Age	Age_Normalized	Salary	Salary_Normalized
0	22	0.000	25000	0.000000
1	25	0.375	30000	0.333333
2	28	0.750	35000	0.666667
3	24	0.250	28000	0.200000
4	30	1.000	40000	1.000000

----- Standardization -----

	Age	Age_Standardized	Salary	Salary_Standardized
0	22	-1.330266	25000	-1.241971
1	25	-0.280056	30000	-0.301084
2	28	0.770154	35000	0.639803
3	24	-0.630126	28000	-0.677439
4	30	1.470294	40000	1.580691

----- Label Encoding -----

	Gender	Gender_Encoded
0	Male	1
1	Female	0
2	Male	1
3	Female	0
4	Male	1

----- One-Hot Encoding -----

	Finance	HR	IT
0	False	True	False
1	False	False	True
2	True	False	False
3	False	False	True
4	False	True	False

----- Final Dataset -----

	Name	Age	Salary	Gender	Department	Age_Normalized	Salary_Normalized	Age_Standardized	Salary_Standardized	Gender_Encoded
0	Amit	22	25000	Male	HR	0.000	0.000000	-1.330266	-1.241971	1
1	Priya	25	30000	Female	IT	0.375	0.333333	-0.280056	-0.301084	0
2	Rahul	28	35000	Male	Finance	0.750	0.666667	0.770154	0.639803	1
3	Sneha	24	28000	Female	IT	0.250	0.200000	-0.630126	-0.677439	0
4	Karan	30	40000	Male	HR	1.000	1.000000	1.470294	1.580691	1